COMPOSITE FUNCTIONS

Total partial derivative theorem for composite function:

If z = f(u, v) where u = ϕ(x, y) and v = ψ(x, y), then the partial derivatives of z are given by

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

Example 1:

Find $\frac{\mathrm{dz}}{\mathrm{dt}}$ given that $z = xy^2 + x^2 y,\ x = at^2,\ y = 2at.$

Solution:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$= \left(y^2 + 2xy\right)\left(2at\right) + \left(2xy + x^2\right)\left(2a\right)$$

$$= 8a^3t^3 + 8a^3t^4 + 8a^3t^3 + 2a^3t^4$$

$$= 16a^3t^3 + 10a^3t^4$$

Example 2:

If $\mathrm{z} = \mathrm{x}^2 + y^2,\ x = \cos uv,\ y = \sin(u+v),$ find $\frac{\partial \mathrm{z}}{\partial \mathrm{u}}$ and $\frac{\partial \mathrm{z}}{\partial \mathrm{v}}$ interms of u and v.

Solution:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$= 2x\left[-v\sin uv\right] + 2y\left[\cos(u+v)\right]$$

$$= -2v\sin uv\ \cos uv + 2\sin(u+v)\cos(u+v)$$

$$= -v\sin 2uv + \sin 2(u+v).$$

Similarly,

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$= -u\sin 2uv + \sin 2(u+v).$$

Example 3:

If z = f(x, y), $x = e^u \sin v,\ y = e^u \cos v$ prove that $x\dfrac{\partial z}{\partial u} + y\dfrac{\partial z}{\partial v} = \left(x^2 + y^2\right)\dfrac{\partial z}{\partial x}$.

Solution:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}$$

$$= \frac{\partial z}{\partial x}\left[e^u \sin v\right] + \frac{\partial z}{\partial y}\left[e^u \cos v\right]$$

Similarly,

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\left[e^u \cos v\right] + \frac{\partial z}{\partial y}\left[-e^u \cos v\right]$$

$$\therefore\ x\frac{\partial z}{\partial u} + y\frac{\partial z}{\partial v} = x\left[x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}\right] + y\left[y\frac{\partial z}{\partial x} - x\frac{\partial z}{\partial y}\right]$$

$$= \left(x^2 + y^2\right)\frac{\partial z}{\partial x}.$$

Example 4:

If $H = f\left(y - z,\ z - x,\ x - y\right)$, prove that $\dfrac{\partial H}{\partial x} + \dfrac{\partial H}{\partial y} + \dfrac{\partial H}{\partial z} = 0.$

Solution:

Let H = f(u, v, w) where $u = y - z,\ v = z - x$ and $w = x - y$.

$$\frac{\partial H}{\partial x} = \frac{\partial H}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial H}{\partial v}\cdot\frac{\partial v}{\partial x} + \frac{\partial H}{\partial w}\cdot\frac{\partial w}{\partial x}.$$

$$= \frac{\partial H}{\partial u}(0) + \frac{\partial H}{\partial v}(-1) + \frac{\partial H}{\partial w}(1).$$

Similarly,

$$\frac{\partial H}{\partial y} = \frac{\partial H}{\partial u}(1) + \frac{\partial H}{\partial v}(0) + \frac{\partial H}{\partial w}(-1).$$

$$\frac{\partial H}{\partial z} = \frac{\partial H}{\partial u}(-1) + \frac{\partial H}{\partial v}(1) + \frac{\partial H}{\partial w}(0).$$

Hence

$$\frac{\partial H}{\partial x}+\frac{\partial H}{\partial y}+\frac{\partial H}{\partial z}=0.$$

Example 5:

If $z=f(x,y)$ and $x=r\cos\theta,\ y=r\sin\theta,$ prove that $\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2=\left(\frac{\partial z}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2.$

Solution:

$$\frac{\partial z}{\partial r}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial r}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial r}.$$

$$=\frac{\partial z}{\partial x}\cdot\cos\theta+\frac{\partial z}{\partial y}\cdot\sin\theta.$$

Similarly,

$$\frac{\partial z}{\partial \theta}=\frac{\partial z}{\partial x}(-r\sin\theta)+\frac{\partial z}{\partial y}(r\cos\theta).$$

Hence,

$$\left(\frac{\partial z}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2=\left(\frac{\partial z}{\partial x}\right)^2\left[\cos^2\theta+\frac{1}{r^2}r^2\sin^2\theta\right]+\left(\frac{\partial z}{\partial y}\right)^2\left[\sin^2\theta+\frac{1}{r^2}r^2\cos^2\theta\right]$$

$$=\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2$$

Exercise

1. If $z=f(x,y)$ and $x=e^u+e^{-v},\ y=e^{-u}-e^v,$ prove that $\frac{\partial z}{\partial u}-\frac{\partial z}{\partial v}=x\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}.$

2. If $x=r\cos\theta,\ y=r\sin\theta,$ find (i) $\frac{\partial^2 r}{\partial x^2}+\frac{\partial^2 r}{\partial y^2}$ (ii) $\frac{\partial^2 \theta}{\partial x^2}+\frac{\partial^2 \theta}{\partial y^2}.$

3. If $z=e^{u-2v}$ and $u=\sin x,\ y=x^2+y^2,$ find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}.$

11. IMPLICIT DIFFERENTIATION:

When $f(x,y)$ is a function of two variables x and y, the equation $f(x,y)=c$ (where c is a constant) enables us to obtain values of y corresponding to values of x. Then we say that y

is an implicit function of x. We assume, for simplicity, that the above equation always defines a unique value of y for each value of x.

Since f is a function of x and y, and y is again a function of x, we can consider f a composite function of x. Then, its total derivative with respect to x is

$$\frac{df}{dx}=\frac{\partial f}{\partial x}\frac{dx}{dx}+\frac{\partial f}{\partial y}\frac{dy}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\frac{dy}{dx}.$$

But since $f(x,y)=c$, the total derivative of f must be identically 0. Thus

$$\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\frac{dy}{dx}=0\Rightarrow\ \frac{dy}{dx}=-\frac{\partial f/\partial x}{\partial y/\partial y}=-\frac{f_x}{f_y}\quad\text{if } f_y\neq 0.$$

Differentiating again with respect to x, considering $\partial f/\partial x$ and $\partial f/\partial y$ as composite functions of x, we get

$$\frac{d^2y}{dx^2}=-\frac{\left(\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y\partial x}\frac{dy}{dx}\right)\frac{\partial f}{\partial y}-\frac{\partial f}{\partial x}\left(\frac{\partial^2 f}{\partial x\partial y}+\frac{\partial^2 f}{\partial y^2}\frac{dy}{dx}\right)}{\left(\frac{\partial f}{\partial y}\right)^2}$$

$$=-\frac{\frac{\partial^2 f}{\partial x^2}\left(\frac{\partial f}{\partial y}\right)^2-2\frac{\partial^2 f}{\partial y\partial x}\ \frac{\partial f}{\partial x}\ \frac{\partial f}{\partial y}+\frac{\partial^2 f}{\partial y^2}\left(\frac{\partial f}{\partial x}\right)^2}{\left(\frac{\partial f}{\partial y}\right)^3}$$

Thus,

$$\frac{dy}{dx}=-\frac{f_x}{f_y}\quad\text{and}\quad\frac{d^2y}{dx^2}=-\frac{f_{xx}\left(f_y\right)^2-2f_{yx}f_xf_y+f_{yy}\left(f_x\right)^2}{(f_y)^3}$$

Example 1: If $x^y=y^x$, find $\frac{dy}{dx}$.

Solution:

$x^y = y^x$

Taking log on both sides, we get

y logx = x logy

f(x, y) = y logx - x logy = 0

$$\therefore \ \frac{\mathrm{dy}}{\mathrm{dx}} = -\frac{f_x}{f_y} = \frac{\frac{y}{x} - \log y}{\log x - \frac{x}{y}} = \frac{y\left(y - x\log y\right)}{x\left(x - y\log x\right)}.$$

Example 2: Prove that if $y^3 - 3ax^2 + x^3 = 0$, then $\frac{d^2y}{dx^2} + \frac{2a^2x^2}{y^5} = 0$.

Solution: Let

$$f(x,y) = y^3 - 3ax^2 + x^3 = 0.$$

We have

$$f_x = -6ax + 3x^2,\ f_y = 3y^2.$$

$$f_{xx} = -6a + 6x,\ f_{xy} = 0, f_{yy} = 6y$$

Therefore,

$$\frac{d^2y}{dx^2} = -\frac{6(x-a)9y^4 + (3x^2 - 6ax)^2 6y}{27y^6} = -2\frac{a^2x^2}{y^5}$$

EXERCISE

1. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, prove that

$$\frac{d^2y}{dx^2} = \frac{abc + 2fgh - af^2 - bg^2 - ch^2}{(bx + by + f)^3}$$

2. If A, B, C are the angles of a triangle such that $\sin^2 A + \sin^2 B + \sin^2 C$ is constant, show that

$$\frac{dA}{dB} = \frac{\tan B - \tan C}{\tan C - \tan A}$$

12. ERRORS AND APPROXIMATIONS

Let $f(x,y)$ be a continuous function of x and y. If δx and δy are the increments of x and y, then the new value of $f(x,y)$ will be $f(x+\delta x, y+\delta y)$. Hence

$$\delta f = f(x+\delta x, y+\delta y) - f(x,y).$$

Expanding $f(x+\delta x, y+\delta y)$ by Taylor's theorem and supposing δx, δy to be small enough that their products, squares, and higher powers can be neglected, we get

$$\delta f = \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y, \text{ approximately.}$$

The value δf is called the error in f due to the errors δx and δy in x and y. $\frac{\delta f}{f}$ is called the relative error in f and $\frac{\delta f}{f}\times 100$ is called the percentage error in f.

Example 1: If $PV^2 = K$ and if the relative errors in P is 0.05 and in V is 0.025 then show that error in K is 10%.

Solution:

$$\frac{\delta P}{P} = 0.05 \text{ and } \frac{\delta V}{V} = 0.025$$

$$PV^2 = K$$

Taking log on both sides of the equation,

$$\log P + 2\log V = \log K$$

$$\delta(\log P) + 2\delta(\log V) = \delta(\log K)$$

$$\frac{1}{P}\delta P + 2\frac{1}{V}\delta V = \frac{1}{K}\delta K \Rightarrow 0.05 + 2(0.025) = \frac{\delta K}{K}$$

$$\frac{\delta K}{K} = 0.1$$

Thus, error is

$$100\times\frac{\delta K}{K} = (0.1)\times 100 = 10\%$$

Example 2: The time T of a complete oscillation of a simple pendulum is given by the formula $T = 2\pi\sqrt{l/g}$. If g is a constant find the error in the calculated value of T due to an error of 3% in the value of l.

Solution:

$$T = 2\pi\sqrt{l/g}$$

Taking log on both sides,

$$\log T = \log 2\pi + \frac{1}{2}\log\frac{l}{g}$$

$$\delta(\log T) = \delta(\log 2\pi) + \delta\left(\frac{1}{2}(\log l - \log g)\right)$$

$$\frac{\delta T}{T} = 0 + \frac{1}{2}\frac{\delta l}{l} - 0$$

$$100\frac{\delta T}{T} = \frac{1}{2}\frac{\delta l}{l}\times 100 = \frac{1}{2}\times 3 = 1.5$$

Thus, the error in T is 1.5%

Example 3: If the sides and angles of a plane triangle ΔABC vary in such a way that its circum-radius remains constant, prove that $\frac{\delta a}{\cos A} + \frac{\delta b}{\cos B} + \frac{\delta c}{\cos C}$, where δa, δb, and δc denote small increments in the sides a, b and c respectively.

Solution:

Let R be the circum-radius of the ΔABC, then

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

$$\Rightarrow 2R\sin A, b = 2R\sin B, c = 2R\sin C$$

Differentiating, we get

$$\delta a = 2R\cos A\,\delta A, \delta b = 2R\cos B\,\delta B, \delta c = 2R\cos C\,\delta C$$

$$\Rightarrow \frac{\delta a}{\cos A} + \frac{\delta b}{\cos B} + \frac{\delta c}{\cos C} = 2R(\delta A + \delta B + \delta C)$$

$$= 2R\delta(A + B + C) = 2R\delta(\pi) = 0.$$